

New Southern Ocean transfer function for subsurface temperature prediction using radiolarian assemblages

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Abstract :

Radiolarians are microzooplankton that produce siliceous shells that preserve well in sediments and allow for paleo-reconstructions. Previous studies have used them for sea surface temperature (SST) reconstructions. However, radiolarians peak in abundances between 100 and 400 m in the Southern Ocean (SO), suggesting that their assemblages are more representative of subsurface conditions. Here, we aim to develop a SO-wide transfer function (TF) for subsurface temperature reconstructions (subST) using the Southern Ocean RAdiolarian Dataset (SORAD), a compilation of data for about 240 radiolarian taxa in 228 surface sediment samples from the SO. This exhaustive dataset has been simplified using common TF criteria to minimize noise and optimize SORAD for subST prediction. Ordination tests and Q-mode Factor Analyses (QFA) were applied to the resulting training dataset, which includes 212 samples and 75 taxa (SORAD212_75). The results suggest that, out of six environmental variables, radiolarian assemblages in SORAD are mainly driven by summer temperatures at 200 m and that the first five factors of the QFA explained over 75% of SORAD212_75 variance. We applied common TF methods, IKM, MAT and Weighted-MAT (WMAT), to two data transformations (relative abundances of species and log-transformed). The six models of modern temperature predictions all show excellent performance over the -2 to 18 °C interval. To compare with previously published results generated with the same method, IKM% was applied to radiolarian census data of three cores from the Atlantic, Indian and Pacific sectors of the SO, thus calculating new subST reconstructions and testing the new TF performance.

Highlights

► Calibration of a new radiolarian transfer function based on SORAD. ► Ordination tests recognize subST at 200 m as a valid target for prediction. ► Q-mode Factor analysis shows main SO features recorded by radiolarian assemblages. ► Three transfer function methods provide excellent calibration performance. ► Comparison of new temperature records from three cores with published results.

Keywords : Southern Ocean, Radiolarian, Transfer function, Subsurface temperature

1. Introduction

Radiolarian polycystines are a siliceous, exclusively marine, holoplankton group (Suzuki and Not, 2015). They synthesize robust shells that are well preserved in sediments, particularly in polar areas where carbonates are usually dissolved (Boltovskoy, 2017) and their assemblages have been showed to vary according to water temperatures (Boltovskoy and Correa, 2016; Hernández-Almeida et al., 2017, and references therein). Thus, for years, transfer functions (TF) have been applied successfully to radiolarian species using diverse regional datasets to estimate paleo sea-surface temperature (SST) and intermediate temperatures (Hays et al., 1976; Abelman et al., 1999; Cortese and Abelman, 2002; Luer et al., 2009; Panitz et al., 2015; Hernández-Almeida et al., 2017; Matsuzaki and Itaki, 2017; Hernández-Almeida et al., 2020).

Sediment trap and plankton net studies in the Atlantic sector of the SO have shown that most radiolarians thrive in subsurface waters (Abelman and Gowing, 1997; Boltovskoy, 2017). In the SO, their maximum flux in the water column corresponds to the maximum deposition of biogenic silica, in the Polar Front Zone (Abelman and Gersonde, 1991; Abelman, 1992). In the Atlantic sector, most of the radiolarian production occurs during the summer season (Abelman and Gersonde, 1991) at water depths of 100-400 m (Abelman and Gowing, 1997; Suzuki and Not, 2015; Boltovskoy, 2017). Recent studies showed that subsurface temperature at 200 m (subST) was identified as the most important variable to explain radiolarian assemblage distribution in high latitude core top samples from Indian and Pacific datasets (Rogers and De Deckker, 2007; Hernández-Almeida et al., 2020). Moreover, ocean temperature is known to control community structure and abundance of radiolarians (Llopis Monferrer et al., 2021), and to regulate many metabolic pathways in biosiliceous organisms, such as biomineralization of silica and growth rate for diatoms (Martin-Jézéquel et al., 2000; Boyd et al., 2010; Boyd et al., 2016). While not many ecological studies exist for radiolarians, it is probable that temperature similarly affects radiolarian metabolism. The lack of results in that respect underlines the need for additional studies on radiolarian ecology, life cycle and metabolism. Tréguer (2014) and Chase et al. (2015) mention that about 1 - 6% of biogenic silica that falls in the water column is preserved in SO sediments, suggesting that thinner radiolarians are preferentially dissolved. Nevertheless, the fact that subST was recognized as the leading variable to explain variability of radiolarian assemblages preserved in surface sediments at high latitudes (Rogers and DeDeckker, 2007; Hernández-Almeida et al., 2020) suggests that radiolarian assemblages are highly representative of past subsurface conditions

near 200 m depth during the summer season. Based on these observations, we aim at establishing a new SO-wide TF for subST predictions. To develop a robust TF, we use the largest and most comprehensive radiolarian dataset in the SO, the Southern Ocean Radiolarian Dataset (SORAD; Lawler et al., 2021). SORAD is a compilation of radiolarian census data in surface sediments derived from the harmonization of three published datasets from the Southern Atlantic, the Southern Indian and the Southern Pacific oceans (Figure 1; Cortese and Abelman, 2002; Rogers and De Deckker, 2007; Cortese and Prebble, 2015; Lawler et al., 2021). It is composed of 228 samples (Figure 1) with data for about 240 radiolarian and phaeodarian taxa (Lawler et al. 2021). It offers new opportunities to better understand regional variability in modern radiolarian assemblages preserved in SO sediments (Lowe et al., 2022), but also to refine the use of radiolarians as a paleo-proxy.

*** Figure 1***

Civel-Mazens et al. (2021a,b) used an unpublished version of SORAD to successfully reconstruct subST at 200 m from radiolarian relative abundances in cores MD12-3396CQ and MD11-3353 from the Kerguelen Plateau region in the Southern Indian Ocean. In this study, we update their methods based on the newer version of SORAD (Lawler et al., 2021). Our aim is to test which of several environmental variables drive radiolarian assemblages in SORAD and to develop a single, robust training dataset that can be used to predict subST in the whole SO. The updated TF is, here, calibrated in three phases based on ordination statistical tests and Q-mode Factor Analyses to test species response to environmental variables and summarize assemblages, respectively. To follow methods used to produce published temperature records in three cores from the Atlantic, Indian and Pacific sectors, we applied IKM on radiolarian relative abundances of these cores (Figure 1, yellow stars), thus evaluating the performance of the new training dataset and TF. Our newly generated subST records are compared to published temperature records for these cores that were estimated using subsets of SORAD (Cortese et al., 2007; Panitz et al., 2015) or an unpublished version of SORAD (Civel-Mazens et al., 2021a). Results show that the dataset can equally predict subST at 200 m in the Southern Atlantic, Southern Indian and South-western Pacific and that modern models obtained with three TF methods all performed exceptionally well.

2. Material and Methods

2.1 From SORAD to training dataset SORAD212_75

The main purpose of SORAD was to provide an open and comprehensive compilation of radiolarian census data in SO samples, including as much available data as possible, with harmonized radiolarian taxonomy (Lawler et al., 2021). When possible, over 300 radiolarian specimens per sample were counted, using the taxonomy concepts described by Petrushevskaya (1967, 1971, 1975), Nigrini and Moore (1979) and Boltovskoy (1998). Currently valid taxonomic concepts have been applied by consulting the online databases at <https://www.marinespecies.org> (World Register of Marine Species, WORMS), <https://www.mikrotax.org/radiolaria/> (Mikrotax), and <https://www.radiolaria.org> (Radiolaria), which represent the most complete sources of information on modern radiolarians, with WORMS being the most authoritative in terms of taxonomic nomenclature. More details on SORAD and its taxonomy are available in Lawler et al. (2021).

The intent of Lawler et al. (2021) was to present raw data in order to give readers the freedom to format the dataset for their own purpose. Here, we follow that intent by aiming to provide a transfer function model that is calibrated and ready-to-run for predictions of subST in the SO. Therefore, the dataset was statistically treated and optimized towards this aim. However, before any statistical analysis of SORAD, data for phaeodarians were removed and abundances of *Actinomma hastatum* Van De Paverd (1995) were added to *Actinomma leptoderma sensu lato* (s.l.), as taxonomy concepts described by Van De Paverd (1995) were never adopted by the radiolarist community.

Usually, the SO is considered to encompass all water bodies south of the subtropical front (STF), from approximately 40°S (Whitworth et al., 1980). The westerly winds that drive the ACC originate from horse latitudes, and western boundary currents that can have a large impact on the SO oceanography are sourced at low to mid latitudes. As such, Lawler et al. (2021) included samples from up to 10°S in SORAD. To better represent SO assemblages in our TF and to correspond with the source region of westerly winds in the Southern Hemisphere, we restricted our database to 30°S and therefore discarded all samples (13 samples) from latitudes north of this limit (Figure 1, black squares). Additionally, three samples with less than 100 radiolarians counted (77 and 44 radiolarian specimens in MD94-113 and MD94-115, respectively) or in which identified radiolarians accounted for less than 50% of the total assemblages (43% of the assemblage in PS2103-2) were removed (Figure 1, black squares).

A Detrended Correspondence Analysis (DCA) and a Canonical Correspondence Analysis (CCA) were applied to the resulting subset of SORAD (Figure 2A and 2B, respectively),

which comprises the relative abundances of 228 taxa in 212 core-top samples (SORAD212_228), along with Summer (January-February-March) environmental variables at 200 m water depth extracted from the World Ocean Atlas 2018 (WOA18). These variables include: Salinity (Sal_200) and Temperature (T_200), both extracted from a 0.25°×0.25° grid (Locarnini et al., 2019; Zweng et al., 2019); Apparent Oxygen Utilization (O_200) and main dissolved macro-nutrients such as Nitrate (N_200), Phosphate (P_200) and Silicate (Si_200), all extracted from a 1°×1° grid (Garcia et al., 2019a,b).

Figure 2

The CCA helped identify which of the tested environmental variables control radiolarian distribution of SORAD212_228. Here, T_200 was the most important explanatory variable, and was positively correlated to the axis CCA1 (see section 3.1). Based on the results of ordination tests applied to SORAD212_228 (Figure 2, Supplementary table 1), some radiolarian species with similar skeletal features, low relative abundances and similar distribution along CCA1 (axis driven by temperatures) were grouped together to increase reproducibility of results (Figure 2, colored dots; Supplementary table 2). As such, the radiolarian groupings were done according to taxa's response to temperatures at 200 m depths, highlighted by the CCA (Figure 2). *Acrosphaera mercurius*, the only outlier in the colonial dot cloud (Figure 2, circled yellow dot), was left out of the colonial group. Following Esper and Gersonde (2014a,b), species near the center of the CCA plot (Figure 2, black dots) were considered as having a low response to the tested environmental variables. These species, that obtained coordinates below an arbitrary threshold of 20% of the maximum length of both axes CCA1 and CCA2 (max length: 1.41, -1.41 and 1.51, -1.64, respectively; 20% of the maximum length: 0.28, -0.28 and 0.30, -0.32, respectively; Figure 2, black rectangle) on the plane (CCA1, CCA2), were discarded (Figure 2, black dots). Following recommendations of Imbrie and Kipp (1971), species that do not reach a minimum relative abundance of 2% in at least one core-top of the dataset, were discarded. Finally, ~15 categories of identification at the genus or family level, which include species with potentially conflicting ecological preferences (e.g., Lithaelidae/Pylonoids, undetermined Spumellaria/Nassellaria, *Dictyophimus* spp., *Zyrgocircus* spp., etc.), were removed. The resulting subset includes a list of 75 taxa or groups of species and 212 samples (Supplementary table 3). Thereafter, all mentions of SORAD refer to this cleaned version, SORAD212_75, which is used for further statistical analysis. A full taxonomic description, including names, synonyms, references, and plates, for most species in SORAD212_75, is presented in the Supplementary Plates. Most pictures

included in these plates were taken using a motorized Leica microscope at the University of Bordeaux. Several pictures of a single specimen were captured at different focus levels and then stacked in Photoshop 2021 to obtain a single image with the whole skeleton in focus.

2.2 Statistical analysis of SORAD212_75

Several DCAs, CCAs, and Q-mode Factor Analyses (QFAs) were applied to SORAD212_75. A first DCA was used on relative abundances to test whether species of SORAD have a unimodal or linear response to environmental variables (Ter Braak and Prentice, 1988). Additionally, the DCA displayed an alternative, corrected distribution of the dataset (Figure 3A). This may be required if an artifact, such as the arch effect (Hill, 1974), occurs in a CCA when using explanatory variables that are strongly correlated to the second axis (Ter Braak and Prentice, 1988). A first CCA was used on relative abundances and environmental variables (extracted from the WOA18) to identify the main driver for SORAD's variance (Figure 3B). Six successive CCAs were applied to relative abundances and each of the six environmental variables individually to perform the λ_1/λ_2 test (also known as CCA1/CA1 test). If λ_1/λ_2 is over 1, the given variable used for this specific CCA run can be reliably predicted with the training dataset (Juggins et al., 2013). Additional runs of DCA and CCA were generated for SORAD212_75 with relative abundances that were log-transformed (base 10 logarithm of relative abundance values + 1; Supplementary Figure 1). A final CCA was carried out on SORAD212_75 with temperatures at 10 m, 100 m, 200 m, 300 m, 400 m, and 500 m as environmental variables, to check whether statistical results and known ecological preferences of radiolarian would match.

Four QFAs were applied to summarize 75 variables (species) in the 212 stations into ten and five factors, with data presented as relative abundances or log-transformed. Here, the log transformation was used to rebalance statistical weights of dominant taxa, in particular *Antarctissa* spp. that reaches up to 80-90% of assemblages in some high latitude stations, compared to much rarer species with relative abundances of a few percentage points. A first QFA exploratory run was performed with 10 varimax factors with relative abundance and log-transformed data. A second run was then applied to both data transformations keeping only the first five factors. This allowed the identification of species responsible for most of the variance in the dataset and to calculate factor scores attributed to each station. These factor scores were interpolated with the software Ocean Data View version 5.6.2 (Schlitzer, 2002) to generate maps of the geographic distributions of the five factors.

The DCA and CCA were run using the vegan package v2.5-7 (Oksanen et al., 2020) in R v4.1.3 (R core team, 2022). The QFAs were applied using PaleoToolBox, with the CABFAC, REGRESS and THREAD routines (Sieger et al., 1999). Second-order polynomial regressions between factors' scores and WOA18 subST was calculated in Grapher.

2.3 Transfer Functions

The Imbrie and Kipp Method (IKM; Imbrie and Kipp, 1971) and the Modern Analogue Technique (MAT; Prell, 1985) are the most commonly used TF methods in SO micropaleontological studies. Most previous studies reconstructing SST from radiolarian assemblages used the IKM (Hays et al., 1976; Abelmann et al., 1999; Cortese and Abelmann, 2002; Cortese et al., 2007; Lüer et al., 2009; Panitz et al., 2015). Recently, Hernández-Almeida et al. (2020) used the weighted-average variation of the MAT, weighted-MAT (WMAT), to reconstruct SST and subST with their Pacific training dataset. Based on an early version of SORAD, Civel-Mazens et al. (2021a) reconstructed subST with IKM while Civel-Mazens et al. (2021b) used WMAT. Here, modern subST was estimated using IKM, MAT and WMAT. The three TF methods were applied to both relative abundances (IKM%, MAT%, WMAT%) and log-transformed data (IKMlog, MATlog, WMATlog) of SORAD212_75, because the latter has been suggested to improve performance in diatom-based TF (Köster et al., 2004).

To test the new TF, subST was reconstructed from data of three cores with published temperature records predicted from radiolarian assemblages. As published temperature records in each of these cores were based on IKM% (Cortese et al., 2007; Panitz et al., 2015; Civel-Mazens et al., 2021a), we chose this method, regardless of its calibration performance, to facilitate comparison with our new results.

The IKM was applied with the PaleoToolBox software (Sieger et al., 1999) by selecting 5 factors. Both MAT and WMAT were applied with the PaleoAnalog free software (Therón et al., 2003) by selecting a maximum of 10 analogs with a dissimilarity cut-off set at 20% using the square-chord distance.

3. Results & Discussion

3.1 Ordination

Ordination analysis was carried out on SORAD212_75 and the six main explanatory variables available in the WOA18 to assess which parameter best explains radiolarian

distribution at 200 m depth in the SO and can reliably be reconstructed in the past. Results between statistical tests applied to percentage and log-transformed data of SORAD did not change by much (Supplementary Figure 1), besides slight discrepancies in the first axis' length measured by DCAs (4.4 SD with DCA% against 3.6 SD with DCAlog) and scores attributed to species, stations, and variables by CCAs (Supplementary Figure 1). As these minor differences do not impact interpretations, only statistical tests applied to the relative abundance data, involving minimal manipulation/transformation, are discussed hereafter.

A DCA expresses the range of length between scores attributed to each station in “standard deviation units (SD)” (Hill and Gauch, 1980). Two stations separated by a gradient length of 4SD would have no species in common (complete species turnover). The DCA applied to SORAD212_75 (Figure 3A), yielded a first axis (DCA1) length of 4.4 SD, suggesting that SORAD has a unimodal response to environmental parameters and a complete turnover of species along DCA1. This indicates that the dataset can predict carefully selected variables and a CCA should be favored to test species' responses to environmental variables (Ter Braak and Prentice, 1988). The proportion of explained variance by the first two constrained axes of the CCA were 60.7 % and 20.7 %, respectively (Figure 3B, Supplementary table 4). In this test, T_200 (subST) was the leading explanatory variable, with a score of 0.9884 along CCA1 (Figure 3B, Supplementary table 4). The final CCA run with temperatures ranging from 10 m to 500 m water depth as environmental variables, attributed the highest score to T_200 confirming that this is the best variable for reconstructions. This extends similar and previous observations in regional core top samples studies (Rogers and DeDeckker, 2007; Hernandez-Almeida et al., 2020) to the whole SO and agrees with radiolarian ecology studies that showed radiolarians to peak in abundance in the SO subsurface waters (Abelmann and Gowing, 1997; Boltovskoy, 2017). It is to be noted that the simplification of SORAD212_228 into SORAD212_75 barely affected the length of DCA1 (4.2 and 4.4, respectively; Supplementary tables 1 and 4) and the scores of environmental variables attributed by the CCA (e.g., 0.9888 and 0.9884 for T_200, respectively; Supplementary tables 1 and 4). This suggests that our data treatment did not alter the representativeness of the dataset and its statistical relationships with tested modern environmental parameters.

The CCA applied to SORAD212_75 showed that subsurface temperature was anticorrelated to dissolved nitrate and phosphate, which obtained scores of -0.9871 and -0.9675 along CCA1, respectively (Figure 3B, Supplementary table 4). This strong anticorrelation between macronutrients and temperature in CCAs applied to modern radiolarian datasets was

previously reported at the regional scale (Cortese and Abelman, 2002; Cortese and Prebble, 2015; Hernández-Almeida et al., 2017; Hernández-Almeida et al., 2020). These opposite patterns arise from coupled oceanographic and biogeochemical processes: macronutrients, injected in Antarctic surface water and subsurface waters by the Antarctic upwelling, are consumed by siliceous producers as these water masses flow northward. As such, macronutrient concentrations decrease whereas water temperatures increase equatorward. The natural anti-correlation of subST with N and P in the SO thus potentially explains the apparent importance of macronutrients as explanatory variables for radiolarian datasets. Although we do not exclude that macronutrients and other parameters may exert a strong effect on radiolarian assemblages at a local scale, it has been shown that water temperature is the main driver for variability in said assemblages at the global ocean scale (Boltovskoy and Correa, 2016; Hernández-Almeida et al., 2017, and references therein). To support that SORAD212_75 can be used for subST reconstructions, we applied the λ_1/λ_2 test proposed by Juggins (2013) which evaluates the predictability of an environmental variable. The λ_1/λ_2 test was applied to all six variables (Table 1), and temperatures at 200 m obtained the highest score of 1.78 (Table 1), largely over the critical value of 1 and, thus, meeting the requirement for a variable to be predictable using a TF (Juggins, 2013). Therefore, our new results, along with similar findings in the literature (Cortese and Abelman, 2002; Rogers and DeDeckker, 2007; Cortese and Prebble, 2015; Hernández-Almeida et al., 2020), provide sufficient evidence that subST is a valid target for estimations from radiolarian fossil assemblages recorded in sediment archives, when SORAD is used as the training dataset.

We note that the distribution of species along the CCA's first two axes displays an arch effect (Figure 3B), suggesting that some of the tested environmental parameters are strongly correlated to CCA2 (second axis). When such an artifact occurs, Ter Braak and Prentice (1988) propose to remove variables correlated to CCA2 or to provide an alternative distribution with a DCA. As a DCA was already used to test how species respond to environmental variables (Figure 3A), we opted against removing environmental parameters for an additional CCA run.

Figures 3

Table 1

3.2 Q-mode Factor Analysis

A Q-mode factor analysis was applied to SORAD212_75 data expressed as percentages (QFA%) and Log-transformed data (QFAllog). The first 10 varimax factors explained a total variance of 87.4% and 85.6%, respectively (Table 2). However, as the 6th factor displayed a large drop in explained variance for both QFA% and QFAllog, only five factors were kept for a second round of runs (Table 2). Using relative abundances or log transformations for ordination analysis had little influence on results, and similarly, both QFA% and QFAllog had close outputs (Table 2, Figure 4, Supplementary Figure 2). As QFAllog explained a slightly higher percentage of total variance (76.9 vs 75.8 %; Table 2) and showed clearer trends in the distribution of Varimax Factor Scores (VFS) plotted against subST of the WOA18 (Figure 4), when compared to QFA% (Supplementary Figure 2), the QFAllog run was retained for interpretation.

The QFAllog helped summarize complex signals in SORAD212_75 and highlighted how geographic distribution of factor scores (and, thus, radiolarian assemblages) fit oceanographic features of the SO. The first three factors (Figures 4-5) together account for 60% of SORAD's variance (Table 2), capturing the contrast in radiolarian assemblages between higher (Factor 1) and lower latitudes (Factors 2 and 3). The first factor in QFAllog explains 27% of variance (Table 2) and presents highest varimax factor scores (VFS) at very low temperatures, between -2 and 2 °C (Figure 4). In this factor, the four taxa with highest Scaled Varimax Factor Scores (SVFS) are *Antarctissa* spp., *Lithomelissa* sp. A, *Cycladophora davisiana* and *Cenosphaera cristata*. The relative abundances of these four taxa increase towards cold temperatures (Supplementary Figure 3). Factor 1 is mainly constrained south of the APF, at latitudes greater than 50 °S, and distributed over the full extent of the Antarctic Zone covered by our database (Figure 5). The second factor explains 19% of variance (Table 2) and has its highest scores at very high temperatures between 10 and 18 °C (Figure 4). In this factor, several taxa have very high SVFS: *Lithelius minor* group, *Lamprocyclas maritima*, *Stylodictya tenuispina*, the colonial group, *Tetrapyle octacantha*, *Axopronum stauraxonium*, *Cenosphaera cristata*. The relative abundances of these taxa increase with temperature (Supplementary Figure 3), except *C. cristata* that reaches maximum abundances at stations in the Weddell Sea. Factor 2 covers the STZ of the Pacific and eastern Indian oceans (Figure 5). This suggests that the distribution of radiolarian assemblages in Factor 2 is mainly driven by the East Australian Current and Indonesian Throughflow, the warmest currents in the study area, due to their strong connections to the South Pacific gyre and the West Pacific Warm Pool, respectively. The third factor explains 14% of SORAD variance (Table 2) and its higher scores are

observed for stations with intermediate temperatures between 6 and 14 °C (Figure 4). The five main species in this factor show disparate relative abundance trends with generally higher abundance in the SAZ or STZ (Supplementary Figure 3). Factor 3's geographic distribution, along with abundance trends of its main species, suggest a strong influence from the Brazil Current (west-Atlantic) and the Agulhas Current system (leakage and return current included) over radiolarian assemblages. Our results suggest that the Agulhas Return Current influence on radiolarian assemblages disappears at 60°E (Figure 5) as the current quickly loses strength, warmth and becomes progressively shallower eastward (Kostianoy et al., 2004; Lutjeharms 2006). Together, Factor 2 and Factor 3 indicate that the water masses associated with Western Boundary Currents and their respective extensions, such as the Indonesian Throughflow, the Agulhas leakage and the Agulhas Return Current, control radiolarian assemblages at low latitudes, as observed in Lowe et al. (2022). The next two factors together account for ~ 17 % of SORAD's total variance and indicate zonal differences in radiolarian assemblages of the SO. The fourth factor explains ~11% of SORAD variance (Table 2) and its highest scores are observed at stations with a temperature range between 2 and 12 °C (Figure 4). Dominating species in this factor, *Cycladophora bicornis*, *Phortidium clevei*, *Actinomma antarcticum* and *Spongopyle osculosa*, all display affinities with waters of the Polar Front Zone (PFZ) or SAZ (Supplementary Figure 3). Highest VFS in this factor are distributed in mid latitudes of the Southern Indian Ocean, downstream of the Kerguelen Plateau (Figure 5). Lowe et al. (2022) showed that radiolarian assemblages downstream of the Kerguelen Plateau were distinct from other regions of the SO as a response from interactions between ACC flow and the Kerguelen Plateau and Islands (see discussion therein). Factor 5 explained substantially less variance than the previous factor (~6%) and VFS absolute values were not as high as in the other factors. The dominant species plot negatively in this factor (Figure 5). They generally thrive in waters over 6 °C except the colder species *Phortidium clevei* and *Cenosphaera cristata* (Supplementary Figure 3). However, we note that the dominant species in this factor have disparate distributions compared to WOA18 temperatures, which makes interpretations arduous (Supplementary Figure 3). Nonetheless, the highest absolute VFS values are observed in stations from the western Pacific (Figure 5) as already observed in the regional study of Cortese and Prebble (2015) who attributed their Factor 4 (similar to our Factor 5) to the SAZ in the Tasman Sea and the Campbell Plateau.

In the QFAllog, *Cenosphaera cristata* was part of the main species in four factors out of five (Factors 1, 2, 3 and 5). This species' relative abundances plotted against WOA18

temperatures reveal that it increases in abundances with temperatures, reaching 10-12% in samples associated with water temperatures of 12-14 °C (Supplementary Figure 3). However, in some samples from the Weddell Sea, associated with water temperatures of 0-2 °C, it reaches up to 40-50% of relative abundance. As such, *C. cristata* has a strong signal in both colder and warmer factors of the QFAlog. Such pattern may arise from the existence of two different varieties of the same species and/or unknown ecological preferences of *C. cristata*. According to Riedel (1958), the former is probable, as the initial description of this species from Haeckel is very brief and not illustrated. As there is currently no evidence for the existence of two different varieties of *C. cristata*, we preferred leaving this species included in our training dataset considering that it has a strong statistical signal and that such a bimodal pattern in its relative abundances could result from unknown ecological preferences.

Table 2

Figure 4 & 5

3.3 Transfer Function

Three methods of prediction were used to build new TF models based on SORAD: IKM, MAT and WMAT. IKM and MAT were chosen because of their common application among TF studies, facilitating comparison with previously published results, and their contrasting calculation methods that are based on different statistical assumptions. While IKM can extrapolate no analogue situations and therefore account for them, MAT interpolates predictions from the information contained in the modern analogs exclusively present in the reference dataset, requiring a training dataset including as many stations as possible. The MAT variation, WMAT, was added here based on the results of Hernández-Almeida et al. (2020). The authors showed that WMAT performed best with their Pacific dataset when compared to the Weighted-Averaged Partial Least Squares method (WAPLS) from Ter Braak and Juggins (1993). In the present study, following the methodology developed for diatoms in Esper and Gersonde (2014a,b), we showed evidence through ordination and literature, that subST are a valid target for reconstructions from radiolarian assemblages using SORAD. Additionally, via the QFA, we showed that radiolarian assemblages varied according to latitude but were also associated with particular water masses/currents of the SO (see 3.2 section above). These results are in good agreement with the recent study of Lowe et al. (2022) which described ecoregionalization of the SO with statistical analysis of SORAD.

Data as relative abundances and their log-transformed values were used to evaluate the predictive power of SORAD212_75. Hereafter, the different runs will be referred to as IKM%, IKMlog, MAT%, MATlog, WMAT% and WMATlog. Linear regressions between the observed and reconstructed temperatures of modern samples, in a range of ~ -2 °C to 18 °C, resulted in r^2 over 0.95 and RMSEP of ~ 1 °C in all cases (Figure 6). IKM% and IKMlog yielded r^2 of 0.95 and 0.96, and RMSEP of 1.2 °C and 1.1 °C, respectively (Figure 6A-B). The same diagnostics for MAT% and MATlog showed r^2 of 0.97 and 0.96 with an RMSEP of 0.9 °C and 0.8 °C, respectively (Figure 6C-D). The WMAT% and WMATlog had a r^2 of 0.97 for both runs and RMSEP of 0.7 °C and 0.6 °C, respectively (Figure 6 E-F). Overall, modern models built for three TF methods on two different transformations of the dataset (% and log) showed excellent performance, supporting the predictive power of SORAD212_75 for subST reconstructions. The results displayed in Figure 6 suggest that both WMAT% and WMATlog performed best (Figure 6E-F), despite very little difference between the six models. We note that log transforms appeared to perform best for IKM whereas relative abundances yielded better results for both iterations of MAT.

Figure 6

3.4 Downcore reconstructions

Cores ODP1089, MD12-3396CQ and Y9 were obtained in the Atlantic, the Indian and the Pacific sectors of the SO, respectively (Figure 1, yellow stars). We here compare our new subST reconstructions based on the SO-wide SORAD212_75 using IKM% with previous temperature records predicted from regional radiolarian-based TF in these three sediment cores (Cortese et al., 2007; Cortese and Prebble, 2015; Civel-Mazens et al., 2021a; Figure 7). The IKM% was chosen to fit the methods used by authors to reconstruct temperatures in these cores. These six temperature records are compared with the δD record of EPICA Dome C ice core (Jouzel et al., 2007; Bazin et al., 2013), used here as a reference for air temperature variability in the Southern Hemisphere over the last 450 kyrs. Before discussing these results, we stress that it is beyond the scope of this study to interpret paleoceanographical implications of the newly generated records. Those implications have been discussed in the original publications referenced above.

The new reconstructions for core ODP1089 show subST values ~ 5 -6 °C lower than their published SST counterpart (Figure 7E-F). This was expected as Cortese et al. (2007) optimized their TF to reconstruct SST while we specifically generated subST. The new subST record better resolves the glacial-interglacial pattern than in the original record when

compared to δD of EPICA Dome C ice cores (Figure 7G). This is particularly true for the older part of the record where the two oldest cycles were not resolved in the previous reconstruction. The better resolution of the glacial-interglacial cyclicity with our new approach may be attributed to the larger database along with the different species roster, which probably allowed to better constrain changes in complex oceanographic processes related to climate variability. The published record for core MD12-3396CQ was based on an unpublished version of SORAD comprising 207 samples and 81 taxa. A similar statistical process was applied to the early version of SORAD used to produce these reconstructions. As such, very little difference is observed between the published and new records (Figure 7C,D). The small discrepancies can be explained by the addition of samples to the dataset and the slightly different species roster. New subST produced for core Y9 (Figure 7A) appears very similar to their SST counterpart, in terms of pattern and absolute value. Panitz et al. (2015) generated SST from the Pacific subset of SORAD published by Cortese and Prebble (2015) from a list of 93 species in 88 samples, most of which are also found in SORAD212_75. This may explain the similar pattern in reconstruction records but not the similar absolute temperature values, whereby subST is expected to be lower than SST. Core Y9 was obtained in a zone where the water column is well mixed because of the shallow flow over the Campbell Plateau (~ 1000 m), causing SST and subST to be similar in absolute values. Additionally, ~ 40 samples out of 60 in core Y9 were affected by a non-analog situation in the study of Panitz et al. (2015), forcing IKM% to interpolate temperatures for most of the record. In our results, less than 15 samples presented a non-analog situation. The reduction of the non-analog situations is probably due to the higher number of samples in the modern database. Our larger database, thus, allows for more robust reconstructions. These new modern models and temperature records highlight two of the main strengths of SORAD212_75: its large geographical coverage and its optimization for subST reconstructions. Generally, more information is contained in SORAD212_75 than in any other radiolarian datasets of the SO and, with the statistical treatment we applied to the original dataset, noise has been minimized. Taken together, our results support that SORAD212_75 is the first radiolarian dataset that can reliably predict subST in three sectors of the SO. Additionally, our study shows that radiolarian microfossils are a very efficient proxy for subST reconstructions in the SO. This is supported by our TF models, which all have exceptionally high coefficient of correlations and low error of prediction.

Figure 7

Conclusion

The predictive strengths of SORAD come from (1) the wide geographical coverage of samples only leaving out the eastern Pacific sector of the SO and (2) the homogenized taxonomy, preventing biases inherent to different taxonomy concepts or counting methodologies. For these reasons, SORAD can ensure robust environmental predictions as shown in this study. The dataset was treated and simplified based on ordination analysis of the global SORAD212_228. Ordination analysis of the resulting subset, SORAD212_75, identified subST at 200 m as the main driver for SORAD assemblages' variability. The QFALog identified the main oceanographic features of the Southern Hemisphere, whose influences were recorded in radiolarian assemblages preserved in SO surface sediments. We calibrated six transfer function models based on different methods, IKM, MAT and WMAT and two data formats, relative abundances and log-transformed relative abundances. Overall, the calibration results indicated that these methods all performed exceptionally well, independently from the data format. IKM% was applied to three sediments cores from different sectors of the SO, showing an improved predictive power of SORAD212_75 over regional databases to reconstruct temperatures in the SO.

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Author statement

MCM did the data treatment, statistical tests, produced results and subST records and wrote the initial version of the manuscript. GC trained MCM in radiolarian taxonomy. GC and XC co-supervised MCM's PhD along with MI and TI. KAL and VL provided extracted data from the WOA18 and substantial advice for the statistical analysis of SORAD. MI and TI provided supervision along with logistical and financial support to MCM's PhD. GC and TI helped reviewing and revise radiolarian taxonomy and plates. All authors interacted on the results, interpretations and successive versions of the manuscript.

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Figure 1: Map of the Southern Ocean with locations of surface sediment samples and the deep-sea sediment cores discussed in this study. Black squares represent core-top samples from the Southern Ocean Radiolarian Dataset (SORAD; Lawler et al., 2021) that were excluded from the TF. Red dots represent samples subsequently included in SORAD212_75 used here to reconstruct subsurface temperatures. Yellow stars show the three cores locations, ODP1089, MD12-3396CQ and Y9, in which subsurface temperatures were reconstructed. Core ODP1089 was obtained during ODP leg 177 (1999). Core MD12-3396CQ was obtained during MD189 Indien sud-2 (2012). Core Y9 was obtained during National Institute of Water and Atmosphere (NIWA) research voyage 3034 (1997). Colored lines represent the Antarctic Circumpolar Current frontal system after Park et al., (2019). Red line: Subantarctic Front (SAF); blue line: Antarctic Polar Front (APF); gray line: Southern Antarctic Circumpolar Current Front (SACCF).

Figure 2: Results of DCA and CCA applied to SORAD212_228. A: DCA results; blue dots: species; brown squares: stations. B: CCA results; brown dots: *Antarctissa denticulata*, *Antarctissa strelkovi*, *Antarctissa* spp.; purple dots: *Anthocyrtidium ophirensense* and *Anthocyrtidium zanguebaricum*; yellow dots: colonial species including category “Collosphaerid”, circled yellow dot: *Acrosphaera mercurius*; orange dots: *Dictyocoryne profunda* and *Hymeniasstrum euclidis*; green dots: *Eucyrtidium* species including taxa “*Eucyrtidium* spp.”; pink dots: *Hexacantium enthacanthum* and *Hexacantium pachydermum*; red dots: *Lampromitra coronata* and *Lampromitra quadricuspsis*; gray dots: *Lithelius minor* and *Lithelius spiralis*; blue dots: other species; black dots: species with low response to environmental variables. The black rectangle shows 20% of the maximum length of CCA1 and CCA2. Black arrows show environmental variables, T_200: temperatures at 200 m; N_200: dissolved nitrates at 200 m; P_200: dissolved phosphates at 200 m; Si_200: dissolved silicates at 200 m; Sal_200: Salinity at 200 m; O_200: apparent oxygen utilization at 200 m.

Figure 3: Results of DCA and CCA applied to SORAD212_75. A: DCA results; blue dots: species; brown squares: stations. B: CCA results; blue dots: taxa. Black arrows show environmental variables, T_200: temperatures at 200 m; N_200: dissolved nitrates at 200 m; P_200: dissolved phosphates at 200 m; Si_200: dissolved silicates at 200 m; Sal_200: Salinity at 200 m; O_200: apparent oxygen utilization at 200 m.

Figure 4: Varimax factor scores (VFS) attributed to each station by the Q-mode Factor Analysis (QFA) with log-transformed data plotted against subsurface temperatures at 200 m (subST) of the World Ocean Atlas 2018 (WOA18; Locarnini et al., 2019). Black curves represent second degree polynomial regressions. Yellow line: SubTropical Front associated with subST of approximately 12 °C (Belkin and Gordon, 1996). Red line: the Subantarctic Front associated with the northernmost position of the 6 °C temperature isotherm at 200 m (Park et al., 2019). Blue line: Antarctic Polar Front associated with the northernmost position of the 2°C temperature isotherm

at 200 m (Park et al., 2019). Gray line: Southern Antarctic Circumpolar Current Front associated with the northernmost position of the 0°C temperatures isotherm at 100-150 m (Park et al., 2019).

Figure 5: Geographic distribution of factor scores interpolated over the Southern Ocean for the five factors generated by the Q-mode Factor Analysis applied to log-transformed data. Taxa that obtained scaled varimax factor scores with absolute values > 2 are considered the most important for the concerned factor. They are displayed with their scores between brackets on the left side of each map. Note that the scales for maps of Factor 4 and Factor 5 have reversed scales.

Figure 6: Linear regressions between extracted temperatures from the World Ocean Atlas 2018 (WOA18; Locarnini et al., 2019) and modern temperatures estimated by the three transfer function methods Imbrie and Kipp Method (IKM), Modern Analog Technique (MAT) and Weighted-Modern Analog Technique (WMAT), applied to SORAD212_75 with input data as relative abundances (%) and log-transformed (log). Blue colored results were generated with IKM; green colored results were generated with MAT; red colored results were generated with WMAT. Squares correspond to results generated from relative abundances (%); dots correspond to results generated from data as log-transforms (log).

Figure 7: Ocean temperature reconstructions in cores ODP1089, MD12-3396CQ and Y9 calculated using IKM with input data as percentages, and compared to the δD record of EDC ice core vs age in thousands of years before present. A) blue thin line: new subsurface temperature (subST) reconstruction in core Y9. B) gray thick line: published sea surface temperature (SST) record in core Y9 (Panitz et al., 2015). C) blue thin line: new subST reconstruction in core MD12-3396CQ. D) gray thick line: published subST record in core MD12-3396CQ (Civel-Mazens et al., 2021a). E) blue thin line: new subST reconstructions in core ODP1089. F) gray thick line: published SST record in core ODP1089 (Cortese et al., 2007). G) brown line: δD measured in EDC ice core with the age model of AICC (Jouzel et al., 2007; Bazin et al., 2013). All reconstructions are presented with age models associated with their respective core.

Table 1: Results of the λ_1/λ_2 test for each of the six variables extracted from the World Ocean Atlas 2018. λ_1 corresponds to the CCA1 eigenvalue; λ_2 corresponds to the CCA2 eigenvalue. Each of these lambda values was calculated by applying a CCA to the dataset and the single concerned variable.

Table 2: Amount of variance explained by each factor of the four Q-mode Factor Analysis runs.

Supplementary Figure 1: Comparison between DCA and CCA results applied to relative abundance data in SORAD212_75 and log-transformed values. A: DCA results on relative abundances. B: CCA results on relative abundances. C: DCA results on log-transformed data. D: CCA results on log-transformed data.

Supplementary Figure 2: Varimax factor scores (VFS) attributed to each station by the Q-mode Factor Analysis (QFA) with data as percentages plotted against subsurface temperatures at 200m of the World Ocean Atlas 2018 (WOA18).

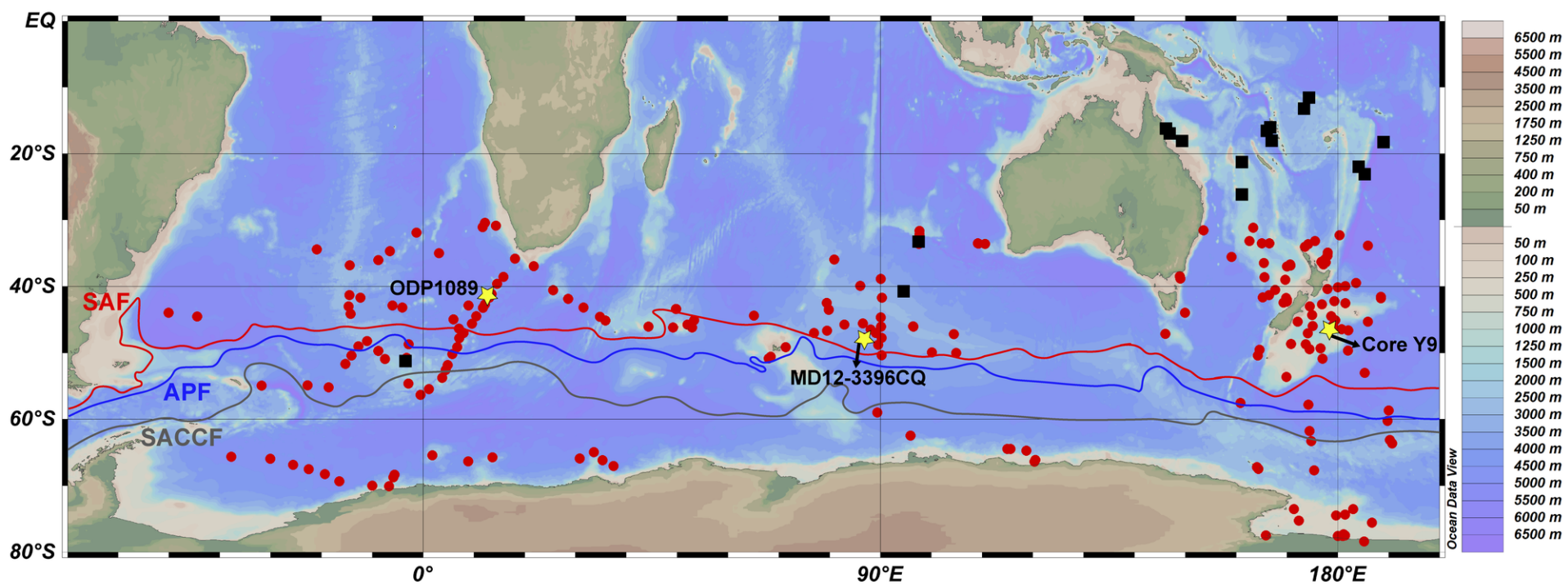
Supplementary Figure 3: Relative abundances of the main species identified in the q-mode factor analysis with log transforms (QFAllog) plotted against temperatures of the World Ocean Atlas 2018 (WOA18). Species that obtained a Scaled Varimax Factor Score (SVFS) > 2 in the QFAllog run with five factors carried out on SORAD212_75 were plotted. Gray plots: Factor 1; orange plots: Factor 2; yellow plots: Factor 3; blue plots: Factor 4; red plots: Factor 5. Yellow line: SubTropical Front associated with subsurface temperatures of approximately 12 °C (Belkin and Gordon, 1996). Red line: the Subantarctic Front associated with the northernmost position of the 6 °C temperature isotherm at 200 m (Park et al., 2019). Blue line: Antarctic Polar Front associated with the northernmost position of the 2°C temperature isotherm at 200 m (Park et al., 2019). Gray line: Southern Antarctic Circumpolar Current Front associated with the northernmost position of the 0°C temperatures isotherm at 100-150 m (Park et al., 2019).

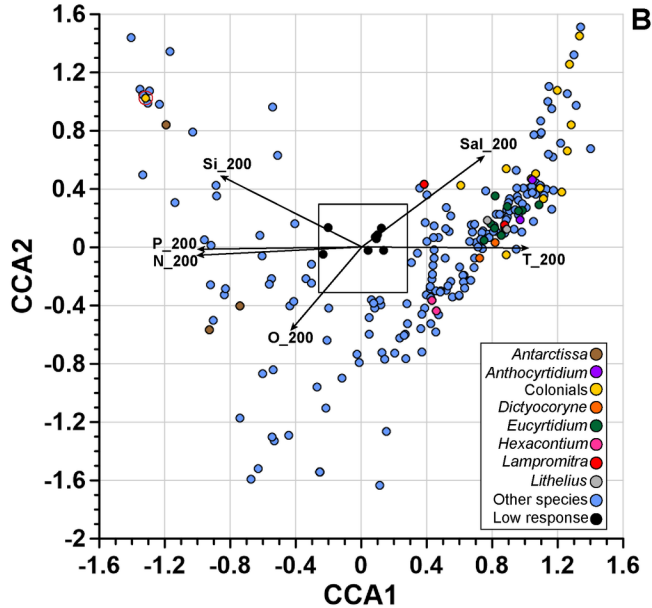
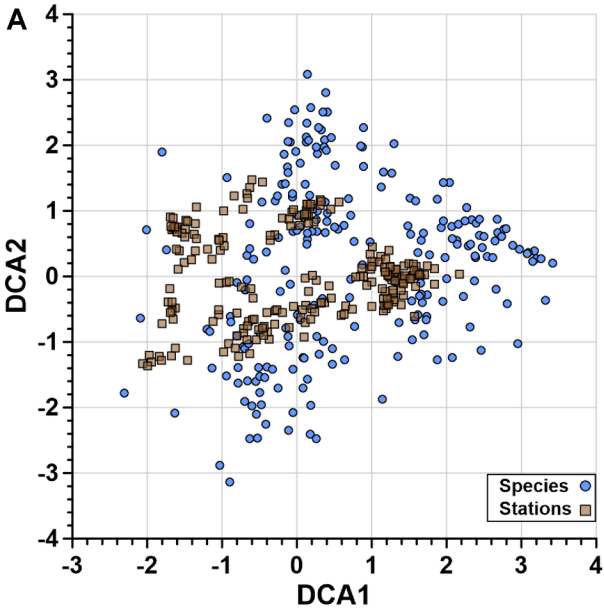
Supplementary Table 1: Results of DCA and CCA applied to species relative abundance data of SORAD212_228.

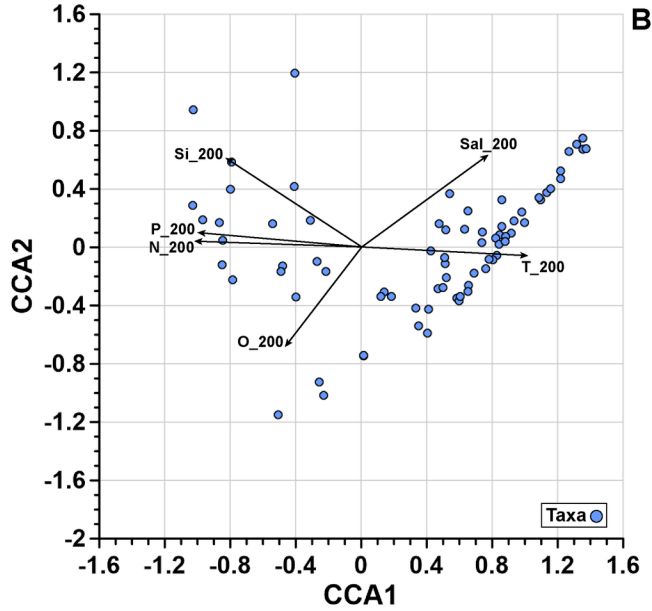
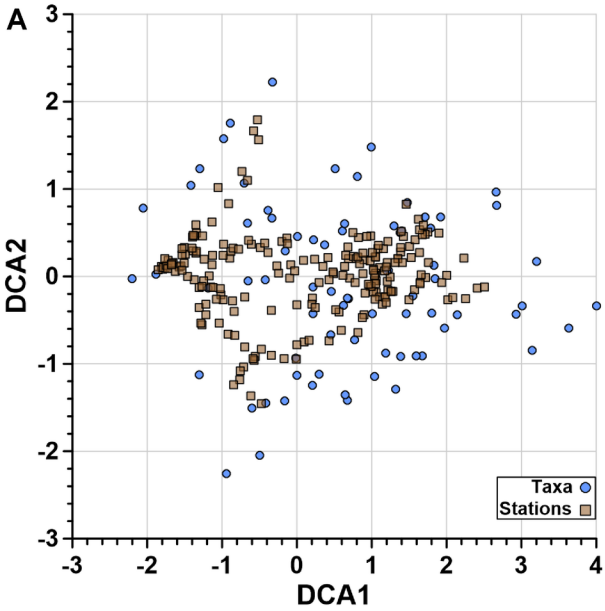
Supplementary Table 2: List of groups of species/taxa that were used to generate SORAD212_75.

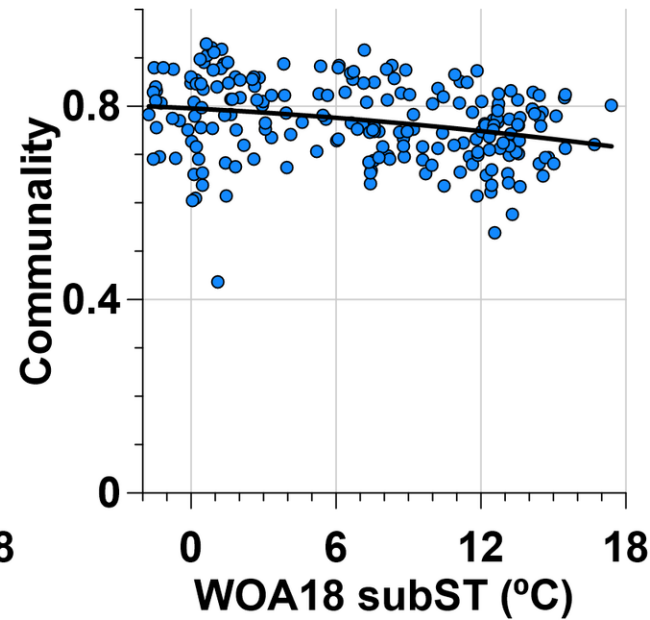
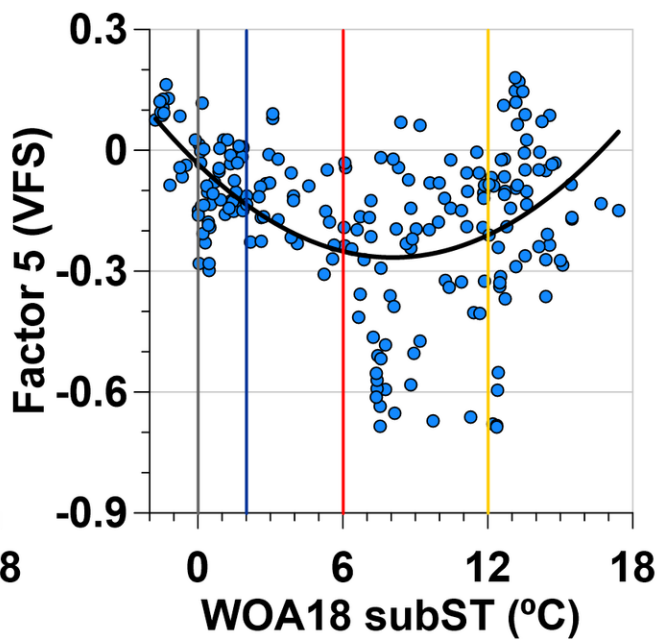
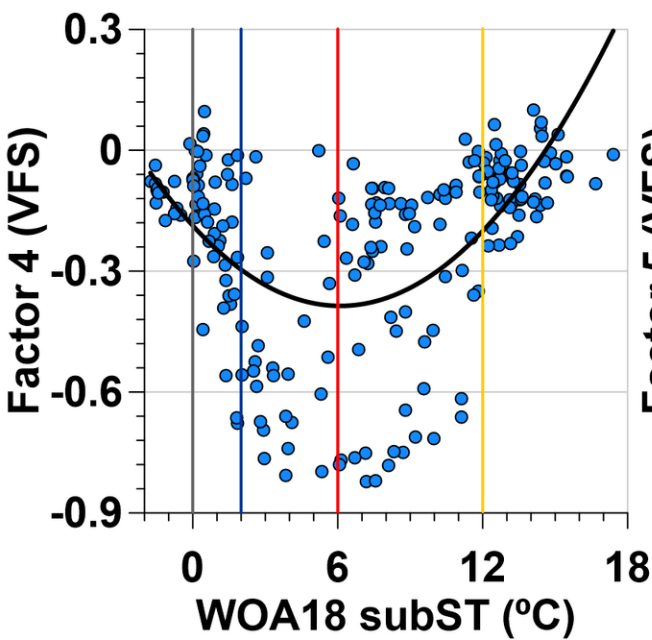
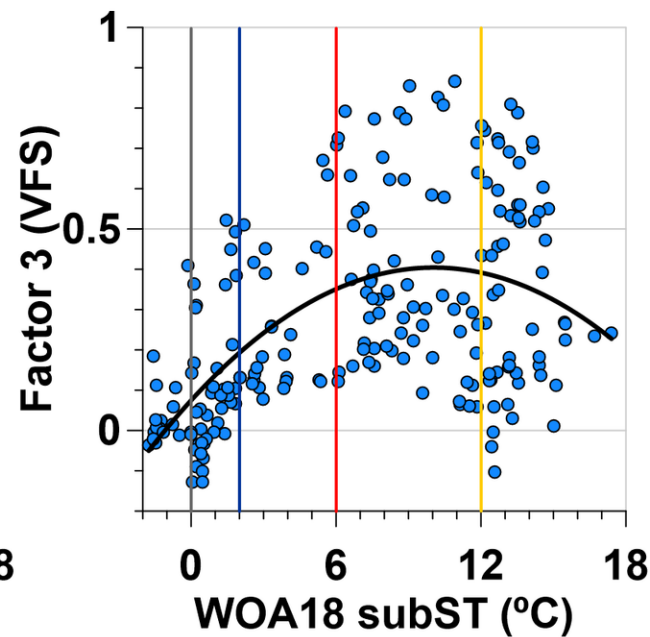
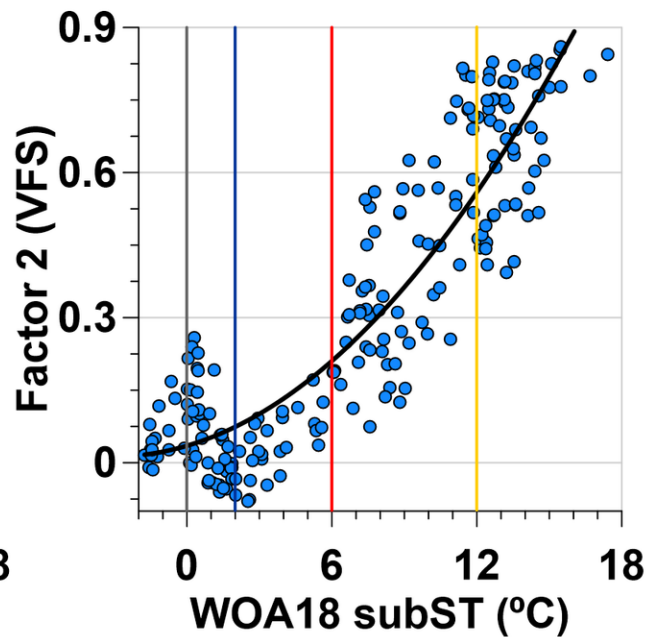
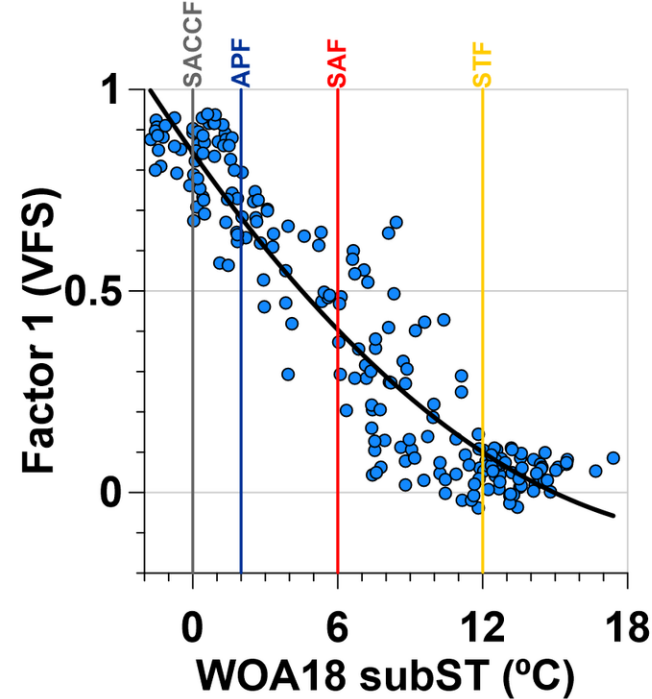
Supplementary Table 3: Training dataset used throughout this study, SORAD212_75.

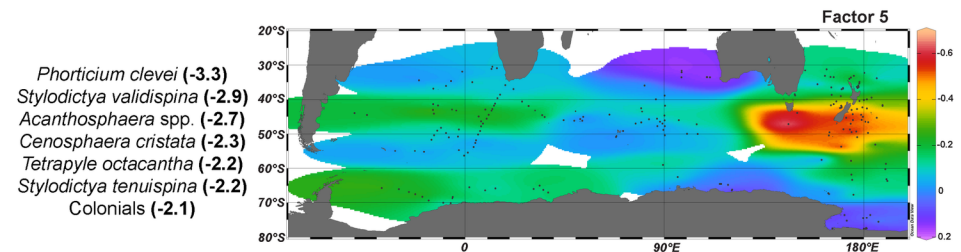
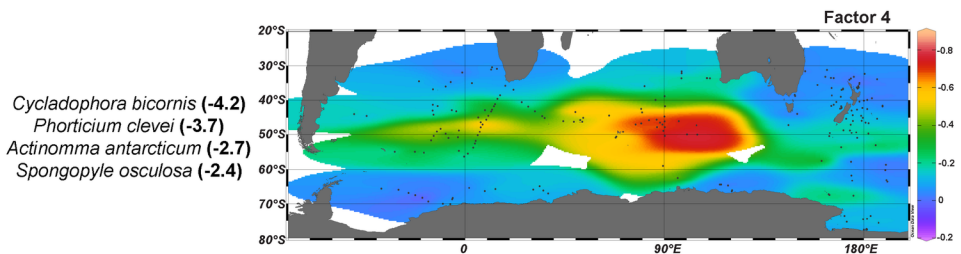
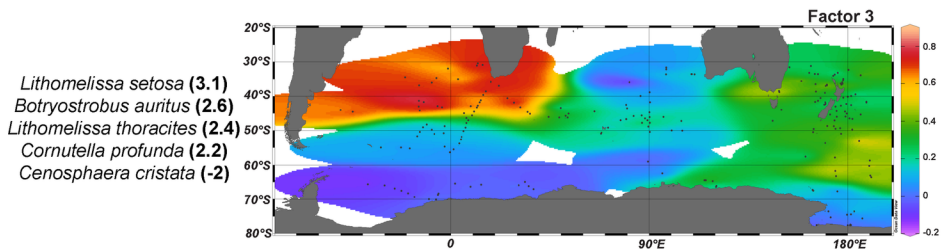
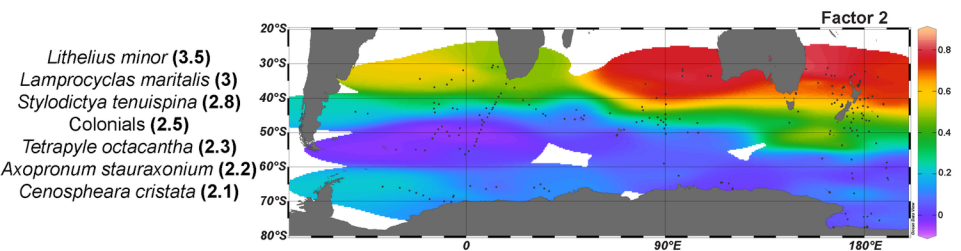
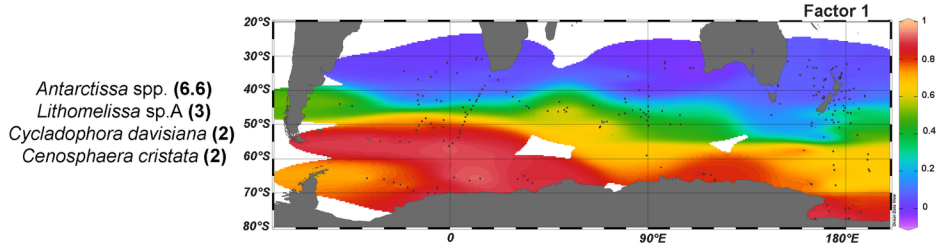
Supplementary Table 4: Results of DCA and CCA applied to SORAD212_75.

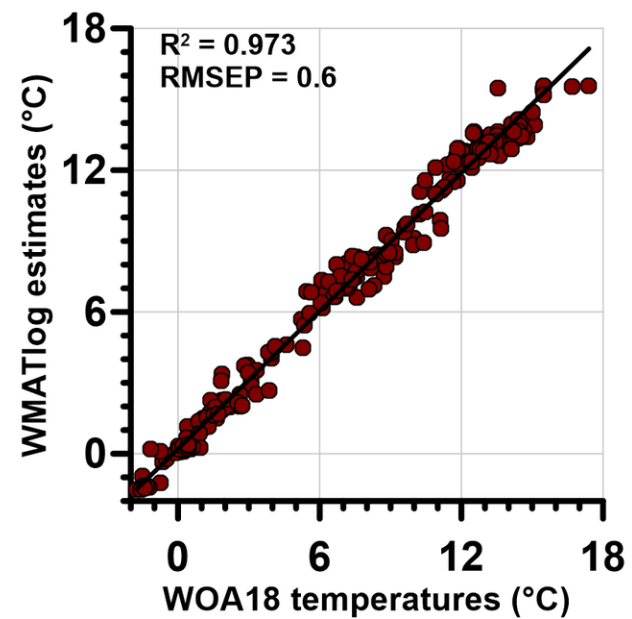
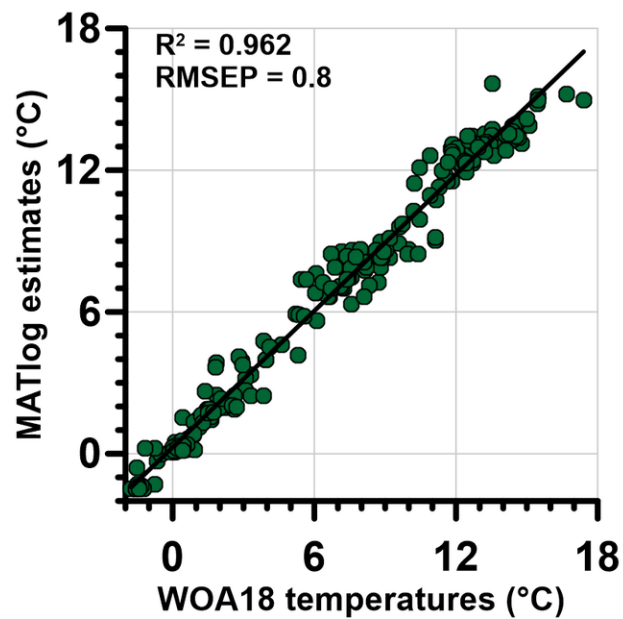
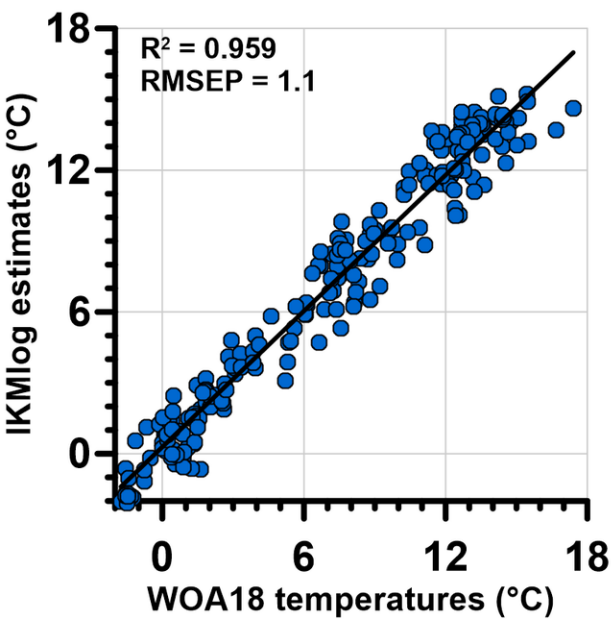
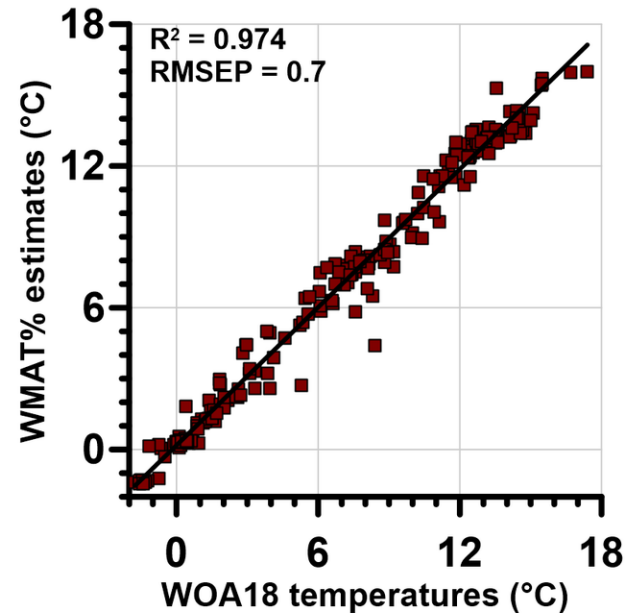
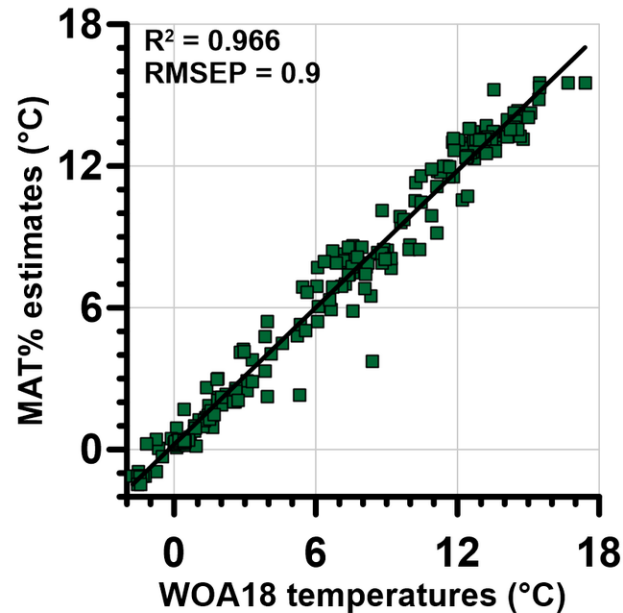
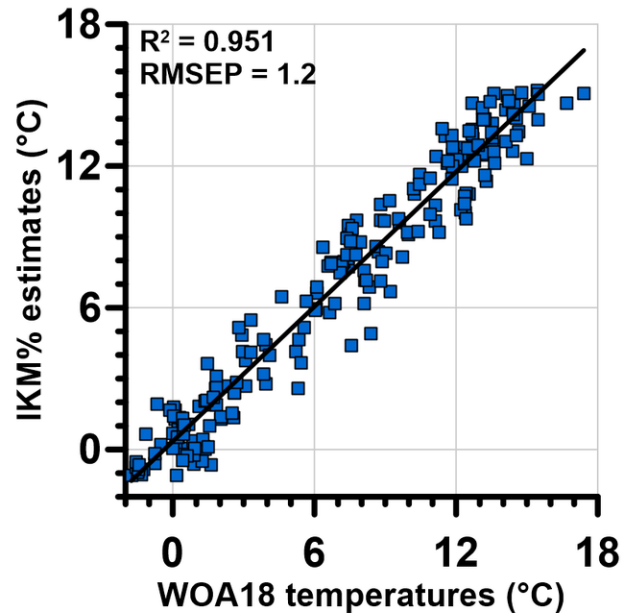


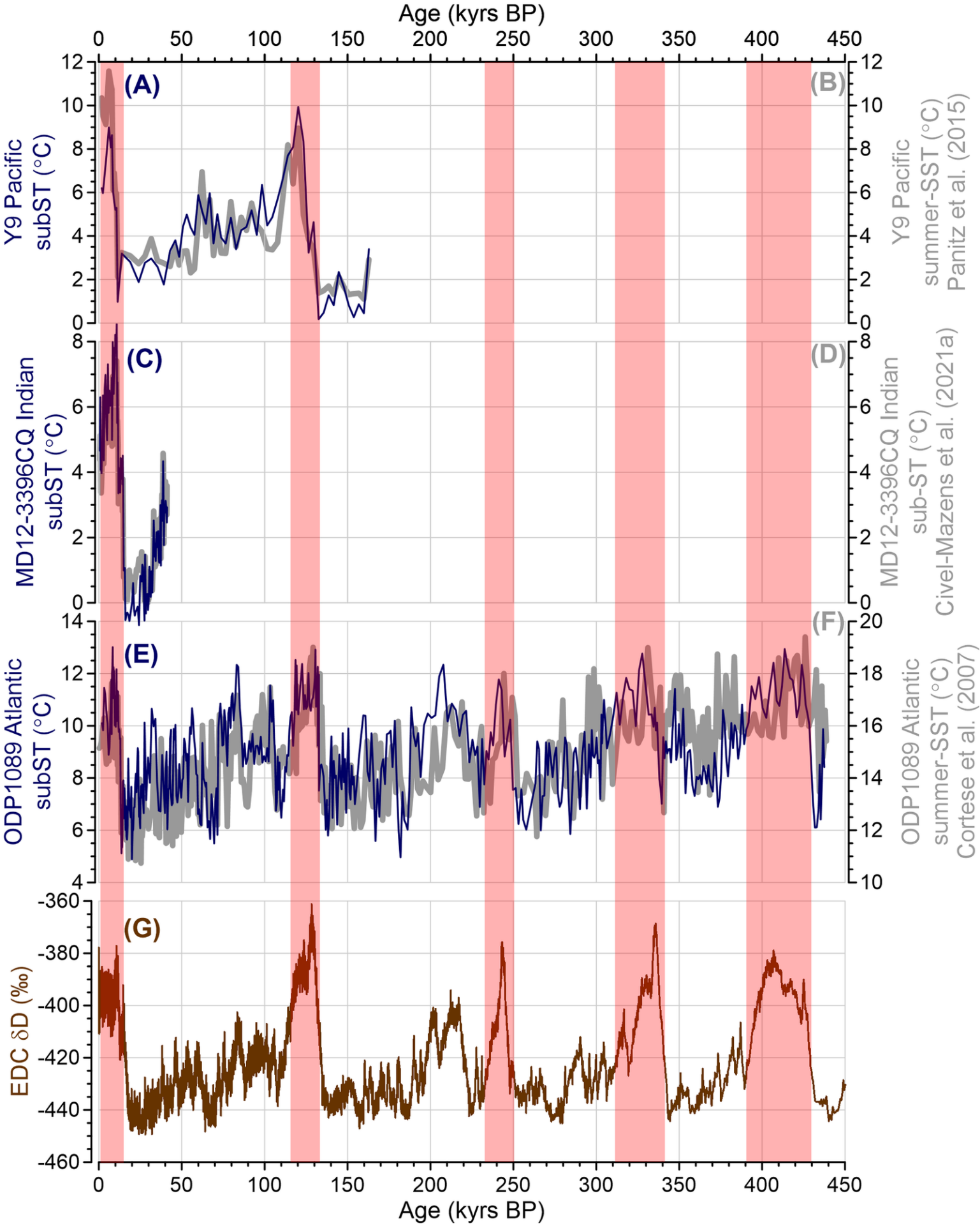












Variable	Code	CCA1	CA1	$\lambda_{CCA}/\lambda_{CA}$
Temperature	T_200	0.5936	0.334	1.77724551
Nitrate	N_200	0.5924	0.3347	1.76994323
Phosphate	P_200	0.5723	0.3386	1.69019492
Silicate	Si_200	0.46	0.4418	1.04119511
Salinity	Sal_200	0.4251	0.4353	0.97656788
Oxygen	O_200	0.22585	0.5604	0.4030157

Method	Variance	Factor 1	Factor 2	Factor 3	Factor 4	Factor 5	Factor 6	Factor 7	Factor 8	Factor 9	Factor 10
QFA% (10 Fact.)	Variance	34.6	10.4	6.4	7.3	3.3	7.7	8.1	4.4	3.2	2.0
	Cumulative variance	34.6	45.0	51.4	58.6	62.0	69.7	77.8	82.1	85.3	87.4
QFALog (10 Fact.)	Variance	26.1	18.0	13.5	10.9	5.7	2.7	3.0	2.1	1.8	1.9
	Cumulative variance	26.1	44.1	57.5	68.4	74.1	76.8	79.8	81.9	83.7	85.6
QFA% (5 Fact.)	Variance	34.7	16.1	11.7	7.6	5.7	N/A	N/A	N/A	N/A	N/A
	Cumulative variance	34.7	50.8	62.4	70.1	75.8	N/A	N/A	N/A	N/A	N/A
QFALog (5 Fact.)	Variance	27.0	19.0	14.0	10.6	6.3	N/A	N/A	N/A	N/A	N/A
	Cumulative variance	27.0	46.0	60.0	70.6	76.9	N/A	N/A	N/A	N/A	N/A